

wages and prices while the new-Keynesin school assumes rigidities in wages and prices.

1.3 BASIC FEATURES OF CLASSICAL THEORY

Classical economics evolved against the perception of ‘mercantilism’. The mercantilists believed that the wealth of nations depended on the stock of bullion (or, gold and silver) and adopted policies that promoted exports and discouraged imports through subsidies and tariffs. Classical economists discarded such ideas and believed that ‘wealth of nations’ depended on real factors. For them, money is merely a medium of exchange. Important features of classical theory are as follows:

- (i) **Microeconomic Issues:** Classical economists dealt mostly with microeconomic issues pertaining to behavior of economic agents such as firms and households. In classical view, a firm maximizes its profits subject to a resource constraint. Similarly, households try to maximize their utilities or economic gains given their budget constraints. Classical economists believed in the optimising tendencies of the market mechanism. According to classical economists, variables such as price level, wage rate and output level should be determined by market forces (supply and demand).
- (ii) **Laissez Faire:** Classical economists believed in the philosophy of ‘laissez faire’, which is a French term meaning ‘leave alone’ or ‘let you do’. According to this view, there should be minimal intervention from the government in business affairs. In fact, Adam Smith suggested that government should confine itself to three main duties, viz., (i) national defense, (ii) administration of justice (law and order), and (iii) establishing and maintaining certain public works (infrastructure, education, etc.).
- (iii) **Invisible Hand:** Adam Smith introduced the concept of the ‘invisible hand’. According to him, the economy will function well if everyone pursues his/ her own interest. According to him, “It is not from the benevolence of the butcher, the brewer, or the baker that we expect our dinner, but from their regard to their own interest”. Individuals pursuing self-interest seem to be led by an ‘invisible hand’ to maximise the general welfare of everyone in the economy. It is not the generosity of a producer in selling a commodity; it is his/her self interest. Similarly, the consumers are not doing a favour to the producer; they are pursuing their own interest in buying the commodity. The philosophy of the invisible hand confined classical economists to the analysis of the behavior of economic agents; they failed to see any conflict between interest of economic agents and that of the economy as a whole.
- (iv) **Continuous Market Clearing:** Classical economists assumed that prices and wage rates are flexible. As you know from microeconomics, equilibrium price is determined at the level where supply and demand are equal. Given the supply and demand curves, if demand is more compared to supply, price will increase. Similarly, if supply is more compared to demand, price will decrease. This principle applies not only to commodities, but also to wage rate. If supply of labour is more than its

demand, there will be a decline in wage rate till supply of labour equates its demand. An implication of the above is that there is no unemployment in the economy – wage rate will decline till all workers are employed. There is no disequilibrium in the market.

- (v) **Perfect Competition:** Classical economists assumed that there is perfect competition in the market so that markets function smoothly. As there is full employment (due to flexibility in wage rate), production is always at the full employment level. An implication of the above is that there is no scope for ups and downs in the level of output. Going by this logic, the classical economists ruled out the possibility of ‘business cycle’.
- (vi) **Say’s Law of Market:** Classical economists believed that production or supply is the key to economic prosperity. Thus, they emphasized more on the supply side of the economy. This approach is summarized very well in the Say’s law, named after the prominent classical economist J B Say. According to J B Say, ‘supply creates its own demand’. Whenever some production takes place, there is a stream of income in the hands of people, which generates demand.

A person should have produced something to sell (say, for example, labour) and thereby earned certain income. Thus, in the classical viewpoint, there is no scope of deficiency of demand in the economy. Therefore, primary concern before an economy is production or supply, not demand.

- (vii) **Neutrality of Money:** According to classical economists, economic growth of an economy is due to increase in the factors of production and technological progress. Money is just a medium of exchange; and it facilitates transactions among economic agents. Thus, increase in money supply does not affect the level of output – it only leads to increase in prices (see quantity theory of money discussed below). They believed that there is dichotomy between monetary variables such as money supply and prices, and real variables such as output and employment. Thus, classical economists stressed the role of real factors in deciding the real variables such as output and employment. As mentioned earlier, classical economists believed in free market systems without government intervention. Determination of output, employment and interest rate in the classical system is given below.

We discuss some of the major issues in the next section.

Check Your Progress 1

- 1) Write down the important features of classical theory.

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2) What are the main features of new-classical economics?

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1.4 DETERMINATION OF OUTPUT AND EMPLOYMENT

The classical theory is based on the following assumptions:

- a) firms and workers are optimisers,
- b) they have perfect knowledge, and
- c) they operate in competitive (perfect) markets.

An implication of the above is that wages and prices are fully flexible. Under perfect competition, the demand for labour for the profit maximising firms is

$$P = W/MPN \quad \dots (1.1)$$

where, P is the product price,

W is nominal wages, and

MPN is the marginal product of labour.

As there is perfect competition, P is equal to Marginal Revenue (MR) which is received from the sale of one unit of output. Here, W/MPN represents the marginal cost of production, that is, the cost of production of an additional unit of output.

Equation (1.1) can be rewritten as

$$MPN = W/P \quad \dots (1.2)$$

(We denote nominal wage by W and real wage by its lower case, i.e., w)

Equation (1.2) indicates that the marginal product of labour is equal to the real wage (W/P). Hence, the labour demand curve in terms of real wage is nothing but the Marginal Product of Labour. The labour demand curve is downward sloping.

$$MPN_d = f(W/P) \quad \dots (1.3)$$

The supply of labour varies positively with the real wage. It assumes that individual labour maximises its utility.

This can be written as

$$N_s = g(W/P) \quad \dots (1.4)$$

where N_s indicates the supply of labour. We explain below in Fig. 1.1 the equilibrium in the labour market. Note that the labour market is in equilibrium at full employment of labour. At the equilibrium point, the supply of labour is equal to demand for labour. In the upper panel, we provide labour supply and labour demand as functions of real wage $\left(\frac{W}{P}\right)$. In the lower panel, we provide labour supply and labour demand as functions of nominal wage. As price level

increases from P_1 to P_2 and then to P_3 , real wage can be maintained at the same level if nominal wage increases to $2W$ and $3W$ respectively. The shifts in the supply curve of labour and demand curve of labour are shown in panel (b) of Fig. 1.1. Notice that there is no change in labour supply, as full employment is ensured (since there is no change in real wage). If there is an increase in output prices, but no increase in nominal wage rate, there will be a decline in real wage. This will lead to a decrease in labour supply.

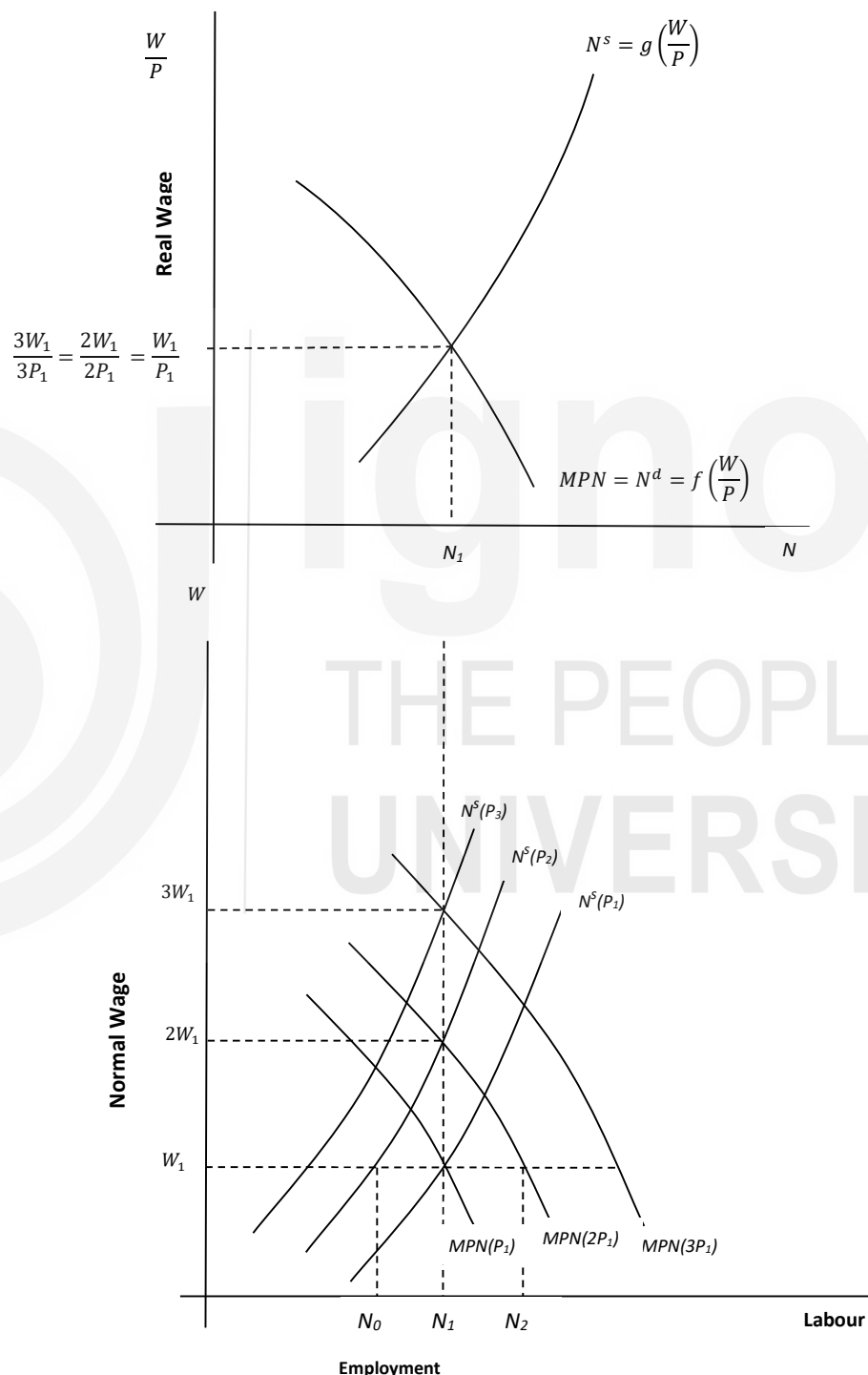


Fig.1.1: Supply of and Demand for Labour

Let us assume that there are only two factors of production, i.e., labour and capital. Output is determined by the production function $Y=F(K,N)$ as given in Fig. 1.2, panel (b). Equilibrium in the labour market (see panel (a) of Fig. 1.2) determines the full employment labour supply and real wage rate. Based on that full employment output is given in panel (b) of Fig. 1.2. As you can see, the economy operates with full employment output (Y_0) and full employment labour supply (N_0).

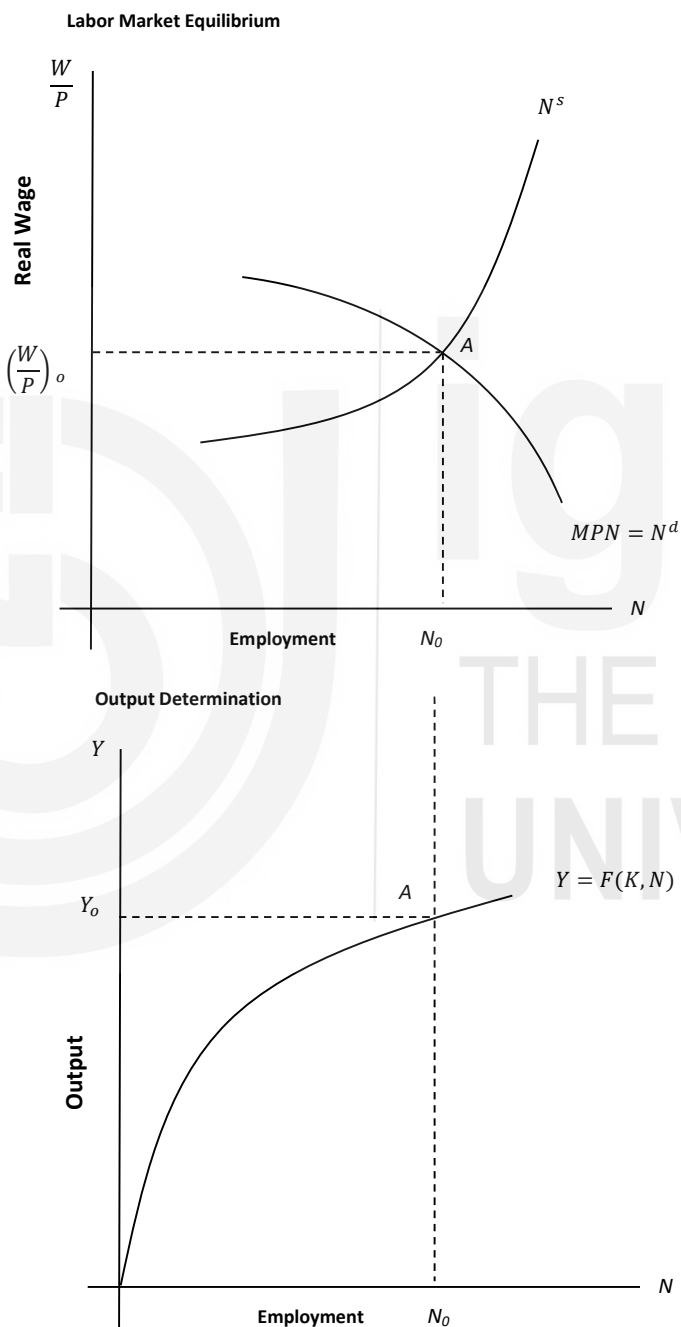


Fig. 1.2: Equilibrium Output and Employment

Since there is full employment in the economy all the time, with a given production function and capital, the Aggregate Supply (AS) is inelastic at full employment level of output. In Fig 1.3 we have shown the AS curve as a vertical straight line. Note that the AS curve will shift if there is a change in the level of technology (production function) or the level of capital.

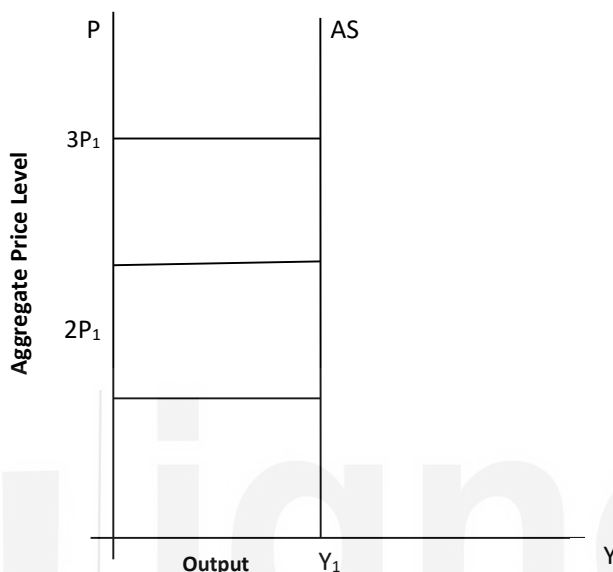


Fig. 1.3: Aggregate Supply Curve

1.5 QUANTITY THEORY OF MONEY

To understand the effects of changes in the stock of money, we need to study the money market equilibrium. Money is a stock variable. Its stock means its quantity at a point of time. Money is an asset demanded by the public for its holding. Money is supplied by the government and the banking system. The interaction of the demand for money and the supply of money create a money market. In this unit, we assume that the monetary authority autonomously determines the supply of money in an economy. Hence, the aggregate demand for money refers to the demand for money by the public. Thus, demand for money means the sum of all the money demanded by individual members of the public, whether households or firms.

There are various theories of demand for money, such as (i) the classical theory of demand for money, or the Quantity Theory of Money (QTM); (ii) the Keynesian theory of demand for money; and (iii) Friedman's restatement of Classical QTM. In this Unit, we will discuss the classical theory of demand for money. It is popularly known as the QTM, which is basically a theory of the price level. Recall that money has four functions; (i) a medium of exchange, (ii) a standard of deferred payment, (iii) a store of value, and (iv) a unit of account. According to classical economists, money is demanded for its function as a medium of exchange. There are several versions of the popular classical QTM. Below we present the transaction version, which is known as the Fisher's equation of exchange:

$$M.V = P.T \quad \dots (1.5)$$

where,

T is number of transactions of average size, and proxy for income level

M is quantity of money supply,

V is velocity of circulation of money, and

P is the average price level.

Classical economists rely heavily on the QTM to establish the theory of demand for money. The QTM says that there is a proportional relationship between the amount of money held by the public and the price level. As per the classical assumption of full employment, the output level is given. Hence, T in equation (1.5) is fixed as a proxy for national income. Further, V is defined as the number of times a rupee moves from one hand to another during a given period. This means that it relies on the public's payment behaviour, which is constant in the long run.

The right-hand side of equation (1.5), PT represents the total amount of money required to facilitate transactions, i.e., the purchase or sale of total output in the economy. The left-hand side of equation (1.5) In the same equation, the left-side term MV is the product of the number of rupees in circulation and the number of times each is used for public payments. Hence, MV represents the total amount of money available for transactions during the given period. Equilibrium is achieved in the system at the point where money demanded (PT) equals money supply (MV).

The equation (1.5) can be rewritten as below,

$$P = \frac{V}{T} M \quad \dots (1.6)$$

As mentioned earlier, the terms V and T are constants in equation (1.6). Thus, the equation indicates that money supply directly affects price level. For instance, if M is increased by four times, price level P will rise by four times. Therefore, classical QTM is known as a theory of price level.

The other approach of classical QTM describes the relationship between the demand for money and 'nominal output' by taking directly real output instead of number of transactions. In equation form, it can be expressed as

$$MV = PY \quad \dots (1.7)$$

where,

M is the money supply

V is the velocity of money that is the number of times the money changes hand.

P is the output price, and

Y is the real output level.

The total stock of money in the economy, measured by money supply times the velocity of circulation (MV), equals the nominal value of transactions or the nominal value of income or output in the economy (PY). The identity given at (1.7) is converted into the QTM under the assumption that Y and V are stable or constant in the short run. With Y and V being constant, the assumption that price level is passive means that P depends on changes in M rather than M depends on changes in P . These assumptions indicate that any short-run decrease (or increase) in M must lead to a proportional decrease (or increase) in P . Relaxation or invalidity of any one of these assumptions would not hold the proportionality relationship of P with M .

According to classical theory, under perfect competition with full employment in the economy, the output level is fixed ($\bar{Y}$). The velocity of circulation is predetermined ($\bar{V}$) and money supply is exogenously given. From equation (1.6), Price level (P) is proportional to the money supply (M).

$$P = (\bar{V} / \bar{Y}) \times M \quad \dots (1.8)$$

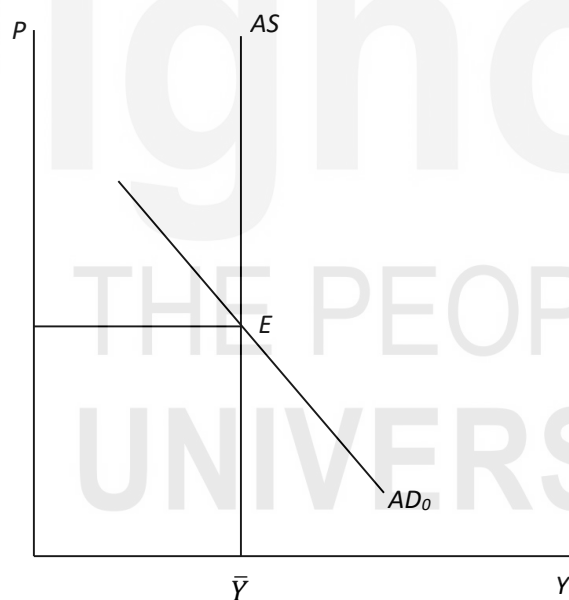


Fig. 1.4: Equilibrium Output and Prices

Thus, the labour market and production function determine the output level while money supply determines the price level in the classical scheme. The quantity theory of the money also gives the aggregate demand curve (see Fig. 1.4). Although the classical theory is accepted by most economists, Keynesian economists and Monetarists have criticized it. They believe that the classical theory fails in the short run when the prices are sticky. Also, it has been established that the velocity of money does not remain constant over

time. Nevertheless, the classical theory can be used to understand and control inflation in the economy.